STA380

TUT 11

STA380 Practice Problems for Quiz 7

4. Derive the log likelihood function of the $Beta(\alpha, \beta)$ distribution. With $n = 10^4$ replicates, use `rbeta(n, shape1 = 3, shape2 = 4)` to generate samples from the Beta distribution. Then, use the Newton-Raphson Method to approximate the MLE of the generated sample.

$\Gamma(\alpha) = (\alpha-1)!$

① Log-likelihood.

$$L(\theta|x) = \left\{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)}\right\}^n \underbrace{\prod_{i=1}^{n} x_i^{(\alpha-1)}} * \underbrace{(1-x_i)^{(\beta-1)}}$$

$$\ell(\theta|x) = n\cdot \underbrace{\log\left(\left\{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\right\}\right)} + (\alpha-1)\sum_{i=1}^{n}\log(x_i) + (\beta-1)\sum_{i=1}^{n}\log(1-x_i)$$

Review:

$\log(a\cdot b)$

$= \log a + \log b$

$\log\left(\Gamma(\alpha+\beta)\cdot\Gamma(\alpha)^{-1}\cdot\Gamma(\beta)^{-1}\right)$

$$= n\left\{\log(\Gamma(\alpha+\beta)) - \log(\Gamma(\alpha)) - \log(\Gamma(\beta))\right\}$$

$$+ (\alpha-1)\sum_{i=1}^{n}\log(x_i) + (\beta-1)\sum_{i=1}^{n}\log(1-x_i)$$

5. Consider $X_i \overset{i.i.d}{\sim} Exponential(\theta)$ ← scale. where $i = 1, 2, \ldots, n$ where $n = 10^4$. Suppose we had all of the realisations except for one of them, let's say X_1. Construct the EM algorithm and code it in R. Use `rexp(n-1, rate = 1/3)` to generate an imcomplete sample.

EM Algorithm :

↓ Expectation ↘ Maximization.

Method is for finding parameter estimates when data has missing value or latent variable.

① E step ⇒ "Fills in" the missing data using current guesses.

② M step ⇒ optimize as if data were complete.

③ Repeat above steps until convergence.

Pdf : $f(x) = \frac{1}{\theta} e^{-\frac{1}{\theta}x}$

① Log-likelihood of ?

$$L(\theta | \underbrace{X_{-1}}_{\text{no } X_1}) = \frac{1}{\theta} e^{-\frac{1}{\theta} \cdot x_2} \cdots \frac{1}{\theta} e^{-\frac{1}{\theta} x_n}$$

$$= \left(\frac{1}{\theta}\right)^{n-1} e^{-\frac{1}{\theta} \sum_{i=2}^{n} x_i}$$

$$\ell(\theta | X_{-1}) = -(n-1)\log(\theta) - \frac{1}{\theta} \sum_{i=2}^{n} x_i$$

$$\frac{d\ell(\theta | X_{-1})}{d\theta} = -\frac{(n-1)}{\theta} + \frac{1}{\theta^2} \sum_{i=2}^{n} x_i \overset{\text{set}}{=} 0$$

times θ^2 for both sides $\Rightarrow -\theta(n-1) + \sum_{i=2}^{n} x_i = 0 \Rightarrow \hat{\theta}^{(0)} = \dfrac{\sum_{i=2}^{n} x_i}{n-1}$

Complete Data log-likelihood.

$$L_c(\theta|x) = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta}\sum_{i=1}^{n} x_i}$$

$$l_c(\theta|x) = -n\ln(\theta) - \frac{1}{\theta}\sum_{i=1}^{n} x_i$$

$$= -n\ln(\theta) - \frac{1}{\theta}x_1 - \frac{1}{\theta}\sum_{i=2}^{n} x_i$$

Expected value.

$X_1 \sim Exp(\theta)$

$E(X_1) = \theta$

$$E\left[l(\theta|x) \,\middle|\, \hat{\theta}^{(k)}, x_{-1}\right]$$

$$= E\left[-n\ln(\theta) - \frac{x_1}{\theta} - \frac{1}{\theta}\sum_{i=2}^{n} x_i \,\middle|\, \hat{\theta}^{(k)}, x_{-1}\right]$$

$$Q(\theta, \hat{\theta}^{(k)}) = -n\ln(\theta) - \frac{\hat{\theta}^{(k)}}{\theta} - \frac{1}{\theta}\sum_{i=2}^{n} x_i.$$

$\Downarrow$

M-step: $$\frac{dQ(\theta, \hat{\theta}^{(k)})}{d\theta} = \frac{-n}{\theta} + \frac{\hat{\theta}^{(k)}}{\theta^2} + \frac{1}{\theta^2}\sum_{i=2}^{n} x_i \overset{set}{=} 0$$

Times $\theta^2 \Rightarrow = -\theta n + \hat{\theta}^{(k)} + \sum_{i=2}^{n} x_i = 0.$

$$\Rightarrow \hat{\theta}^{k+1} = \frac{\sum_{i=2}^{n} x_i + \hat{\theta}^{(k)}}{n}$$